

ABCD Stats Workshop- Module 3

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Agenda

1. Brief overview of inferential statistics
2. Walkthrough templates for:
   a) T-test
   b) ANOVA
   c) Simple Linear Regression
3. Practice updating templates
4. Wrap-Up + Plan for Module 4

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Inferential Statistics

How do we predict who is going to go to start using cannabis prior to age 16?

Inferential Stats is one of many tools we can use to guide our decision making

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Descriptive vs. Inferential Statistics

Descriptive Statistics

- Mean, Median, Mode
- Standard Deviation
- Interquartile Range
- Graphs and Figures

Describe Data

Inferential Statistics

- T-Test
- ANOVA
- Regression
- Correlation

Quantify Relationships

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Terms to Reference

Terms

Independent Variable (IV)

- Predictor Variable

Dependent Variable (DV)

- Outcome Variable

Continuous Variable

- On a Scale

Categorical Variable

- Discrete Categories

Inferential Statistics

- T-Test
- ANOVA
- Regression
- Correlation

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Inferential Statistics- When to Use What

Inferential Statistics

T-Test

- Continuous DV
- 2-Level Categorical IV

ANOVA

- Continuous DV
- +2-Level Categorical IV

Regression/Correlation

- Continuous DV and Continuous IV

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Inferential Statistics- R Templates

T-Test

ANOVA

Regression

Something you can adapt and come back to later

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Inferential Statistics- R Data

Get the files

Three files. Put all three in the `data/` folder of your RStudio Project — the same place `abcd-synthetic.csv` went in Module 0.

data.RDS | data_dictionary.csv | data_dictionary_levels.csv

File	What it holds
`data.RDS`	The data itself — one row per participant per visit
`data_dictionary.csv`	One row per **variable**: what it is, who reported it, which instrument it came from
`data_dictionary_levels.csv`	One row per **level** of every coded variable: `1` means Male, `2` means Female, and so on

A real ABCD release arrives exactly like this: data in one place, meaning in another. Learning to work with the pair is part of the point.

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WARNING

NEVER UPLOAD DATA TO AN LLM

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